

PQE

PQE AI Assistant

Procurement Quality Engineering — AI-Assisted Supplier Audit Platform

User Manual · Version 0.2.0 · Prototype Release

<https://pqe-ai-assistant.vercel.app>

Table of Contents

1. 1. Overview and Purpose
2. 2. Getting Started
3. 3. Core Audit Workflow
4. 4. Checklist Templates
5. 5. Creating an Audit
6. 6. Audit Hub
7. 7. Supplier Self-Assessment
8. 8. Auditor Onsite Verification
9. 9. Findings Log
10. 10. Corrective Action Requests (CARs)
11. 11. PPAP Evidence Review
12. 12. Vertical Evidence Trace
13. 13. Supplier Qualification Decision
14. 14. Audit Report
15. 15. AI Onsite Tools
16. 16. Supplier Risk Dashboard
17. 17. Manufacturing Knowledge
18. 18. AI Features and Controls
19. 19. Settings, Backup and Reset
20. 20. Storage and Data
21. 21. Security and Privacy
22. 22. Frequently Asked Questions

1. Overview and Purpose

PQE AI Assistant is a mobile-first web application designed to support Procurement Quality Engineers, Supplier Quality Engineers, and manufacturing process auditors throughout the complete supplier qualification and onsite audit lifecycle.

The application combines a structured audit workflow with AI assistance (powered by OpenAI GPT-4o and Whisper) to help auditors prepare more effectively, identify risks faster, and produce professional audit records — while keeping all final decisions firmly in human hands.

⚠ PROTOTYPE VERSION — All data is stored in your browser's IndexedDB only. Data is not encrypted at rest and will be lost if you clear your browser data. This application is not approved for storing real production supplier data, confidential drawings, or personally identifiable information until a server-side database is deployed.

Target Users

- **Lead Auditors / PQEs** — plan, execute, and close supplier audits
- **Supplier Quality Engineers** — verify corrective actions and manage CAR follow-up
- **Supplier Contacts** — complete pre-audit self-assessments (via the Supplier Assessment page)
- **Quality Managers** — review the Supplier Risk Dashboard and audit history

2. Getting Started

Accessing the App

- 1 Open **https://pqe-ai-assistant.vercel.app** in any modern browser (Chrome, Safari, Firefox, Edge).
- 2 The app works on mobile and desktop. On mobile, add it to your home screen for a native-app feel.
- 3 No login or account is required for the prototype version.

First-time Setup

- 1 Go to **Settings** (navigation bar) and enter your name in the *Auditor Preferences* field. Click **Save**.
- 2 Your name will be auto-filled as Lead Auditor on new audits and as the verifying auditor on approvals.
- 3 Go to **Checklists** and import your Excel audit questionnaire to create your first checklist template.

Tip: The app works offline once loaded. All audit data is stored locally in your browser — no internet connection is required for data entry, only for AI features.

3. Core Audit Workflow

The recommended workflow follows these stages in order:



PPAP / Trace → Generate Report → Qualification Decision

Stage	Who	Page
Import Excel checklist	Auditor	Checklists
Create audit record	Auditor	Audits → New Audit
Send self-assessment to supplier	Auditor shares the URL	Audit Hub → Supplier Assessment
Supplier completes self-assessment	Supplier contact	Supplier Assessment
Auditor onsite verification	Auditor	Auditor Verification
Record findings	Auditor	Findings Log
Raise and track CARs	Auditor / Supplier	CARs
PPAP evidence review	Auditor	PPAP Review
Evidence trace	Auditor	Evidence Trace
Generate and approve report	Auditor	Report
Record qualification decision	Lead Auditor	Qualification

4. Checklist Templates

Checklists are imported from Excel workbooks (.xlsx) and stored as controlled templates. The original workbook is preserved read-only in your browser storage.

Importing a Checklist

- 1 Navigate to **Checklists** in the navigation bar.
- 2 Click **Import Excel Checklist**.
- 3 Select **.xlsx** file. The file must be structured with columns for reference, question text, guidance, max your score, and mandatory flag. (See Excel format below.)
- 4 Enter a **Checklist Name** and **Revision** number.
- 5 Click **Import**. The checklist is parsed and stored. The original file is preserved.

Expected Excel Format

Column A	Column B	Column C	Column D	Column E	Column F
Section title (merged row)	—	—	—	—	—
Reference (e.g. 4.1.2)	Question text	Guidance notes	Max score	Scoring basis	Mandatory (Y/N)

Rows where Column A contains a section heading (no reference number format) are treated as section dividers.

5. Creating an Audit

- 1 Go to **Audits** → click **+ New Audit**.
- 2 Select the **Checklist Template** to use for this audit.
- 3 Fill in **Supplier Name**, **Site / Location**, and **Supplier Contact**.
- 4 Select **Audit Type**: Qualification, Process, Line, Investigation, or CAR Verification.
- 5 Set **Audit Dates** (one or more days).
- 6 Enter the **Lead Auditor** name and any additional **Team Members**.
- 7 Add a **Scope** description — this is used by AI features for context.
- 8 Click **Create Audit**. You are taken to the Audit Hub.

6. Audit Hub

The Audit Hub is the central page for a specific audit. From here you can navigate to all workflow stages and access AI tools.

Sections

Section	Description
Audit Info	Status, lead auditor, dates, checklist revision, scope
Audit Status	Manually advance status: DRAFT → IN PROGRESS → PENDING APPROVAL → CLOSED
Workflow Steps	Quick links to: Supplier Assessment, Verification, Findings, CARs, Report
Onsite AI Tools	Voice Recording, Document OCR, Drawing Analysis, PPAP Review, Evidence Trace, Qualification
✦ AI Audit Prep	AI-generated risk-based preparation plan AI
✦ AI Daily Summary	End-of-day summary based on current progress AI
✦ AI Agenda & Opening Notes	Day-by-day agenda and opening meeting briefing AI
Agenda	Free-text agenda field set during audit creation

7. Supplier Self-Assessment

The supplier (or the auditor on behalf of the supplier) fills in a conformance status, score, response text, and supporting evidence for each checklist question.

Steps

- 1 From the Audit Hub, click **Supplier Self-Assessment** .
- 2 Select the form language: English, Español, Deutsch, Français, or 中文 using the language toggle at the top.
- 3 For each question, click to expand **Conformance Status** : Conforming, Minor NC, Major NC, Not Applicable, or Not Assessed.
- 4 Enter a score (if the checklist has scoring), a written response, and any comments.
- 5 Attach supporting evidence files (photos, PDFs) using the **Upload Evidence** button.
- 6 Click **Save Response** for each question. Progress is shown in the bar at the top.

Supplier responses are kept completely separate from auditor verdicts. A supplier marking a question "Conforming" does not auto-approve it. The auditor must independently verify each item.

8. Auditor Onsite Verification

This is the core onsite audit activity. The auditor reviews each checklist item independently of the supplier response and records their own verdict based on objective evidence.

Steps

- 1 From the Audit Hub, click **Auditor Verification** .
- 2 For each question, click to expand. The supplier's response is shown in a blue read-only panel for reference.
- 3 Optionally click **Get AI Verification Guidance** for suggested follow-up questions and evidence to request. AI
- 4 Select your **Auditor Verdict** : Conforms, Minor NC, Major NC, Not Verifiable, or Not Applicable.
- 5 Enter a score (if scored checklist) and detailed **Auditor Notes** recording objective evidence observed.
- 6 Attach photos and documents as evidence using the **Evidence** upload.
- 7 Check the **I confirm** approval checkbox to mark the item as auditor-approved.
- 8 Click **Save Verification** .
- 9 When all items are assessed, click **Proceed to Findings** .

Critical control: The approval checkbox (*isApproved*) can only be set by explicitly checking the box. AI suggestions have absolutely no effect on this field. Unchecking reverts to unapproved regardless of AI content.

Verdicts Explained

Verdict	Meaning
Conforms	Objective evidence confirms the requirement is met
Minor NC	Partial non-conformance — does not prevent use but requires correction
Major NC	Significant non-conformance — requires corrective action before approval
Not Verifiable	Insufficient evidence available during the audit to verify
Not Applicable	The requirement does not apply to this supplier / process

9. Findings Log

Findings are formal records of non-conformances, observations, and improvement opportunities identified during the audit.

Creating a Finding

- 1 Go to **Findings** in the navigation (or via Audit Hub → Findings).
- 2 Select the relevant audit from the dropdown.
- 3 Click **+ New Finding** .
- 4 Select the **Classification** : Major NC, Minor NC, Observation, or OFI.
- 5 Enter a **Title** and **Description** .
- 6 Optionally click **Get AI Finding Suggestion** to get a draft finding statement. 
- 7 Check **Raise a CAR** if a corrective action is required — a CAR record is created automatically.
- 8 Click **Create Finding** .

Approving a Finding

Findings are created as **Draft**. To make a finding official, expand it and click **Approve Finding**. This sets `isAuditorApproved = true` and records the approving auditor. AI cannot approve findings.

10. Corrective Action Requests (CARs)

CARs follow a full 8D problem-solving workflow from containment through to auditor-verified closure.

8D Status Flow

OPEN → CONTAINMENT → ROOT CAUSE → CORRECTIVE ACTION → EFFECTIVENESS → CLOSED

Managing a CAR

- Go to **CARs** and select the audit.
- Expand the CAR. Fill in **Owner** (supplier contact responsible) and **Due Date**.
- Fill in each 8D field: D1–D3 Containment, D4 Root Cause, D5–D6 Corrective Action, D7 Effectiveness.
- Upload effectiveness evidence files.
- Click **Save Updates** to save changes, or **Advance to Next Step** to move to the next 8D stage.
- When the **EFFECTIVENESS** stage and you have verified the **✓ Auditor** . A confirmation CAR is at actions are effective, click **Verified Closed** dialog appears.

Critical control: `isAuditorVerifiedClosed` can only be set by the explicit confirmation dialog. AI cannot close CARs. Past-due CARs are automatically flagged OVERDUE and shown on the Dashboard.

11. PPAP Evidence Review

The PPAP page provides a structured review of all 18 PPAP elements. Access it from the Audit Hub → **PPAP Review**.

PPAP Elements Covered

#	Element	#	Element
1	Design Records	10	Material / Performance Test Results
2	Authorised Engineering Change	11	Initial Process Studies (Cpk)
3	Customer Engineering Approval	12	Qualified Laboratory Documentation
4	Design FMEA	13	Appearance Approval Report
5	Process Flow Diagram	14	Sample Production Parts
6	Process FMEA	15	Master Sample
7	Control Plan	16	Checking Aids
8	Measurement System Analysis	17	Customer-Specific Requirements
9	Dimensional Results	18	Part Submission Warrant (PSW)

For each element, set the status to **Submitted / Reviewed / Accepted / Rejected / Not Required**, add auditor notes, and attach evidence. Accepting an element requires a confirmation dialog. The PSW (element 18) shows a banner when accepted — confirming PPAP submission is complete.

12. Vertical Evidence Trace

The Evidence Trace page allows the auditor to verify traceability from a CTF (Critical-to-Function) characteristic on a drawing through to the inspection result — a complete vertical chain.

Trace Chain Steps

1. **Drawing** — CTF characteristic with balloon number, tolerance, and datum
2. **PFMEA** — Characteristic listed with failure mode, severity, occurrence, detection
3. **Control Plan** — Method, frequency, reaction plan, responsible party
4. **Work Instruction** — Operator instruction references the characteristic
5. **Measurement System** — Calibrated equipment with GR&R $\leq 10\%$
6. **Inspection Result** — Actual data showing conformance ($Cpk \geq 1.33$ / 1.67 for CTF)

You can create multiple characteristics (chains) for the same audit. Each link can be Verified, Gap Found, Not Applicable, or Not Checked. Verifying a link requires confirmation. Evidence can be attached to each individual link in the chain.

13. Supplier Qualification Decision

The Qualification page records the formal APPROVE / CONDITIONAL / REJECT decision. It is accessible from the Audit Hub → **Qualification**.

Mandatory prerequisites before a decision can be recorded:

- Audit report must be generated
- Report must be auditor-approved
- Recommended: no open major findings, no open CARs

AI cannot make, suggest, or influence the qualification decision.

Decision Options

Decision	Meaning
✓ Approved	Supplier meets all requirements. Approved for production supply.
△ Conditionally Approved	Approved subject to conditions being met and verified. Record conditions in the text field.
✗ Rejected	Supplier does not meet requirements. Not approved for production supply.
⊕ Deferred	Decision deferred pending further information or re-audit.

Clicking any decision button opens a confirmation dialog showing the supplier name, decision, auditor name, and date before recording.

14. Audit Report

The Report page generates a print-ready HTML audit report and records the auditor's formal approval.

Generating the Report

- 1 From the Audit Hub, click **Generate Report**.
- 2 Select a **Qualification Recommendation**.
- 3 Write the **Conclusion** — summarise findings and basis for recommendation.
- 4 Optionally edit the **Confidentiality Notice**.
- 5 Click  **Generate & Preview Report**. The report opens in a new tab.
- 6 In the new tab, use **File → Print → Save as PDF** to save the report.

Approving the Report

- 1 After generating, click **✓ Approve Report (Auditor)** on the report settings page.
- 2 A confirmation dialog shows your name and asks you to confirm this represents your professional judgement.
- 3 Click Proceed. The report is marked Auditor Approved with your name and timestamp.

- 4 Once approved, the Qualification page prerequisites are met.

15. AI Onsite Tools

All AI tools are accessed from the **Onsite AI Tools** section of the Audit Hub.



Voice Recording

Record audit conversations. Transcribed by OpenAI Whisper. Saved as TRANSCRIPT evidence. Requires explicit consent confirmation before recording starts.



Document OCR

Photograph a supplier document. GPT-4o vision extracts content, checks for missing fields, flags concerns, and suggests follow-up questions.



Drawing Analysis

Photograph a technical drawing. AI identifies CTF characteristics, GD&T callouts, process risks, and required verification documents.



PPAP Review

Structured 18-element PPAP checklist with status tracking, notes, and evidence upload for each element.



Evidence Trace

Vertical traceability chain from drawing characteristic through PFMEA, Control Plan, Work Instruction, MSA, and Inspection Result.



Qualification

Formal APPROVE / CONDITIONAL / REJECT decision page with prerequisites check and confirmation dialog.

Maximum image size for OCR and Drawing Analysis: 10 MB per photo upload (3 MB after base64 encoding for the AI API). For large drawings, take multiple photos of different sections.

16. Supplier Risk Dashboard

The Supplier Risk Dashboard (**Suppliers** in the navigation) aggregates all audit data by supplier and displays a risk profile.

Risk Score Calculation

Risk Level	Condition
HIGH	Any open major findings OR any overdue CARs
MEDIUM	Reserved for future scoring rules
LOW	No major findings and no overdue CARs

Click any supplier card to expand and view the full audit history with dates, audit type, lead auditor, and status. Click **Open** → to go directly to an audit.

17. Manufacturing Knowledge

The Manufacturing Knowledge page (**Manufacturing** in the navigation) provides process-specific reference

information for use during audits.

Nine process modules are available:

Process	Key Focus Areas
CNC Machining	Tool wear, dimensional accuracy, Cpk, CMM verification
Sheet Metal Stamping	Die wear, springback, burr height, material certs
Die / Investment Casting	Porosity (X-ray), alloy certs, process parameters
Plastic Injection Moulding	Material drying, shrinkage, weld lines, regrind control
Welding / Fabrication	WPS/WQR qualification, NDT, filler material traceability
Electroplating / Surface Treatment	Bath chemistry, coating thickness, salt spray, embrittlement bake
Painting / Powder Coating	Pre-treatment control, DFT measurement, adhesion testing
Thermal Processing	TUS/SAT records, AMS 2750, NADCAP, batch traceability
Assembly	Part identity control, torque verification, poka-yoke testing

For each process, five information tabs are available: **Key Risks**, **CTF Characteristics**, **Documents to Request**, **Audit Questions**, and **Common Non-Conformances**.

18. AI Features and Controls

What AI Can Do

- Generate risk-based audit preparation plans
- Suggest follow-up questions and evidence requests for checklist items
- Draft finding statements based on auditor notes and verdict
- Generate end-of-day audit summaries
- Generate day-by-day agenda and opening meeting notes
- Extract and analyse content from photographed documents (OCR)
- Identify CTF characteristics from technical drawing photos
- Transcribe recorded audio of audit conversations (Whisper)

What AI Cannot Do — Ever

Action	Why
Approve a verification item	<code>isApproved</code> requires explicit auditor checkbox
Approve a finding	<code>isAuditorApproved</code> requires explicit auditor button
Close a CAR	<code>isAuditorVerifiedClosed</code> requires explicit confirmation dialog
Approve the audit report	<code>isAuditorApproved</code> requires explicit auditor button
Make a qualification decision	Decision requires prerequisites + auditor confirmation
Accept a PPAP element	Each element requires explicit auditor confirmation
Verify a trace link	Each link requires explicit auditor confirmation

AI Labelling

Every AI-generated output is displayed in an amber box clearly labelled **◆ AI SUGGESTION — NOT AUDITOR APPROVED**. An expandable disclaimer explains the AI's limitations. AI suggestions are never automatically written into any form field.

Models Used

Feature	Model
Audit prep, verification guidance, finding suggestion, daily summary, agenda	GPT-4o-mini
Document OCR, Drawing Analysis	GPT-4o (vision)
Voice transcription	Whisper-1

API Key: AI features require an OpenAI API key set as `OPENAI_API_KEY` in the Vercel environment variables. Without it, AI buttons show a "not configured" message and the rest of the app works normally.

19. Settings, Backup and Reset

Auditor Preferences

Enter your name in Settings → Auditor Preferences. This auto-fills auditor fields on new audits and approvals.

Scoring Thresholds

Set pass/conditional/fail percentage thresholds in Settings → Scoring Thresholds. These are stored in localStorage and used as reference — the qualification decision still requires explicit auditor sign-off.

Exporting a Backup

- 1 Go to Settings → Export Audit (JSON Backup).
- 2 Select the audit from the dropdown.
- 3 Click **Export JSON** . A `.json` file is downloaded to your device.

Photo and document blobs are NOT included in JSON exports (too large). Only structured data (checklist, responses, verifications, findings, CARs, report) is exported. Re-attach evidence files manually after importing on a new device.

Importing a Backup

- 1 Go to Settings → Import Audit from Backup.
- 2 Click **Choose Backup File** and select a previously exported `.json` file.
- 3 If an audit with the same ID exists, it will be overwritten.

Factory Reset

Settings → Factory Reset permanently deletes ALL data in this browser: audits, checklists, findings, CARs, evidence, and reports. You must type **RESET** in the confirmation prompt to proceed. Export all audits before resetting.

20. Storage and Data

Storage Location	What is Stored	Limit
IndexedDB (browser)	All audit records, checklists, evidence metadata, photo and document blobs	~50–500 MB depending on browser/device
localStorage (browser)	Auditor name, current audit ID, scoring thresholds, UI preferences only	~5 MB
Vercel server	Nothing — app is stateless; server only proxies AI API calls	—
OpenAI	Text prompts processed and discarded; audio processed and discarded	—

Data loss risk: Browser data can be lost if you clear browser storage, use private/incognito mode, or switch devices. Always export a JSON backup after completing significant audit work. Photo evidence cannot be recovered from a JSON backup.

21. Security and Privacy

Control	Implementation
OpenAI API key	Stored in Vercel environment variables only — never sent to the browser
AI route	Server-side only (/api/ai/suggest , /api/ai/transcribe)
Input sanitisation	All context fields truncated and sanitised before AI prompt construction
Data at rest	IndexedDB is not encrypted in prototype — do not store real confidential data
HTTPS	Enforced by Vercel for all connections
Security headers	X-Frame-Options DENY, X-Content-Type-Options, Referrer-Policy, Permissions-Policy
Camera / microphone	Permissions-Policy restricts to same-origin only

22. Frequently Asked Questions

AI features show "not configured" — what do I do?

Add your [OPENAI_API_KEY](#) in the Vercel dashboard under Environment Variables and redeploy. See the setup guide for step-by-step instructions.

I cleared my browser data and lost all my audits.

IndexedDB data is cleared when you clear browser storage. Always export JSON backups regularly. If you had a backup, restore it via Settings → Import. Photo evidence cannot be recovered.

Can two people use the app at the same time?

Not in the prototype — each user has their own browser storage. Real-time collaboration requires a server-side

database (future version).

The app shows a "Prototype" banner — is this normal?

Yes. The banner is a reminder that this is prototype storage only, not a production quality record system. It cannot be dismissed permanently in the current version.

Can I use the app on my phone without internet?

Yes — once the app has loaded, a Service Worker caches the app shell so it works offline. AI features (text suggestions, OCR, transcription) require an internet connection. All audit data entry works offline.

The Excel import failed — what format does the file need to be?

The importer reads the first worksheet. Section rows are detected by having no numeric reference in the first column. Each question row should have: reference, question text, guidance, max score (number or blank), scoring basis (text), mandatory flag (Y or blank). Use the existing checklist template as a reference.

How do I print the audit report to PDF?

From the Report page, click **Generate & Preview Report**. The report opens in a new tab. Use **Ctrl+P** (or Cmd+P on Mac) → change destination to **Save as PDF** → Save.

How do I print this manual to PDF?

Open <https://pqe-ai-assistant.vercel.app/manual.html> in your browser, then use **Ctrl+P / Cmd+P** → **Save as PDF**.